Disjoint Set Union (DSU) (Easy)

Time Complexity

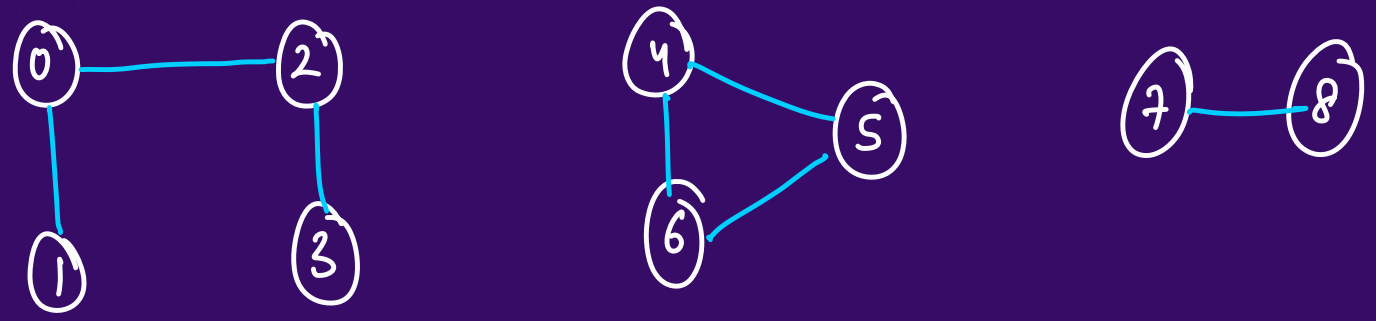
Undirected Graph → Cycle detect

→ No. of connected components

→ Minimum spanning tree

Disjoint Set Union (DSU)

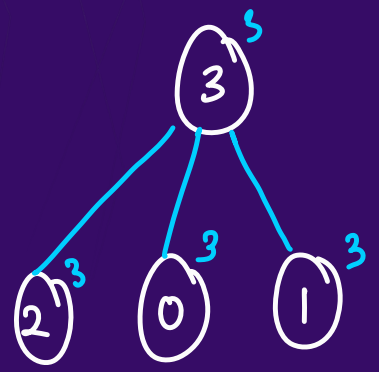
Adjacency Matrix , Adj List , Edge List



edges = { {0, 1}, {0, 2}, {2, 3}, {4, 6}, {4, 5}, {5, 6}, {7, 8} }

Disjoint Set Union (DSU)

edges = { {0, 1}, {2, 3}, {0, 2}, {4, 6}, {5, 4}, {6, 5}, {7, 8} }



find(a) → leader of 'a'

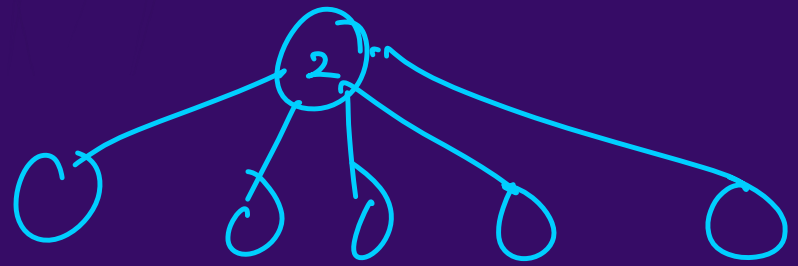
union(a,b) → ek component me le aayega

Disjoint Set Union (DSU)

Optimisation

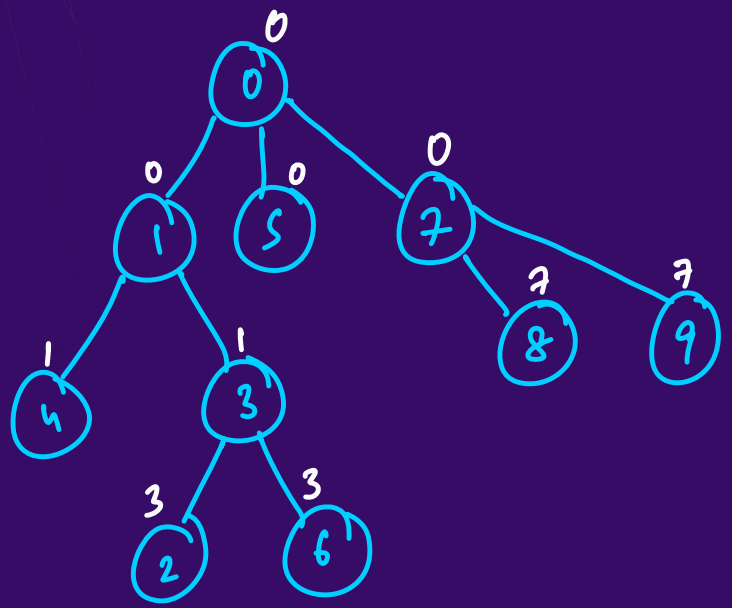
Worst Case

	0	1	2	3	4	5
par	3	4	2	2	0	1



Disjoint Set Union (DSU)

Optimisation



union (9, 3)

7 0

parent[7] = 0 ✓

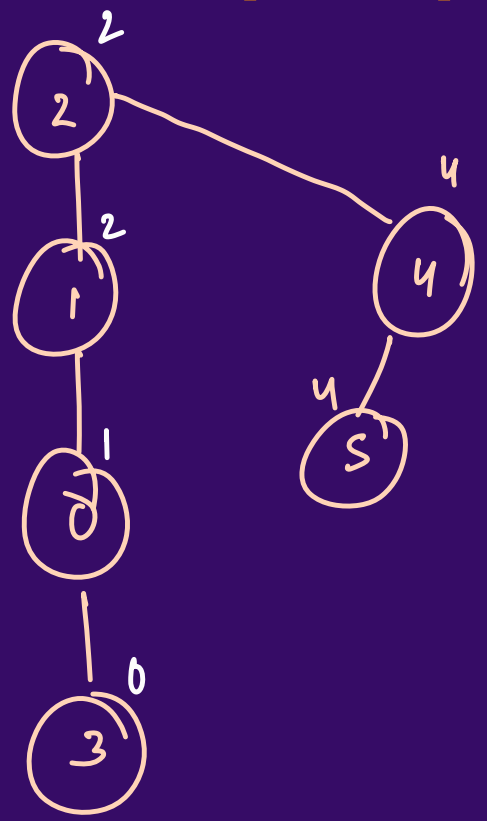
or

parent[0] = 7 ✗

Union by size

Disjoint Set Union (DSU)

Optimisation



union (4, 3)

4 2

Disjoint Set Union (DSU)

Optimisation

Balanced tree v/s Skewed tree v/s two-level tree

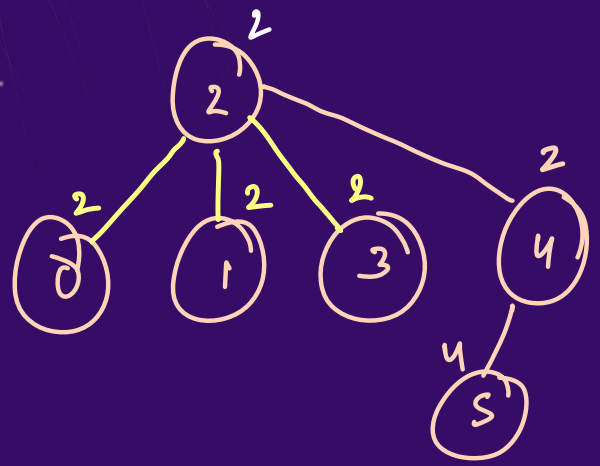
↓ less no. of levels ↓ find(a) ⇒ O(logn)

↓ highest number of levels ↓ find(a) → O(n) Time

↓ find(a) ⇒ T.C. = O(1)

Disjoint Set Union (DSU)

Path Compression



find(3)
↓ ↑2
find(0)
↓ ↑2
find(1)
↓ ↑2
find(2)

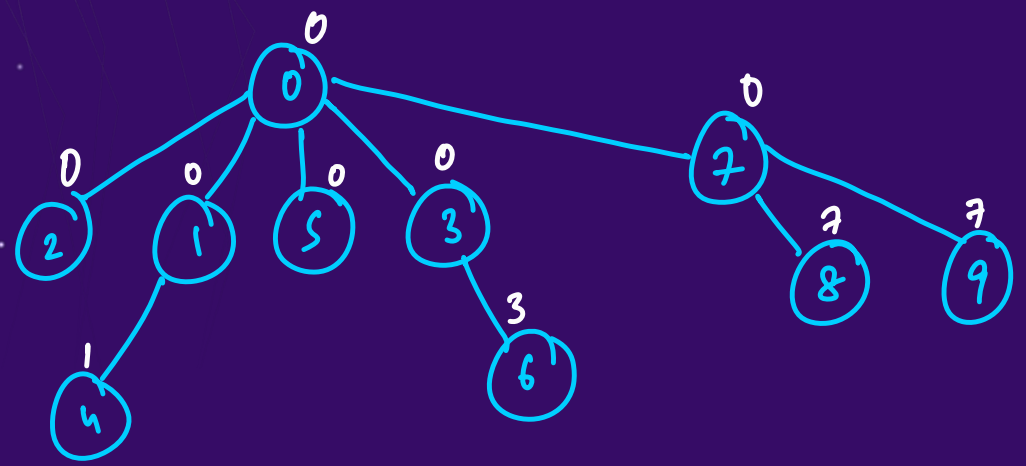
leadent
↑
recursion

union(3, 5)
↓ ↓
2 2

find → path compression

Disjoint Set Union (DSU)

Path Compression



union (2, 9)
↓ ↓
0 7

DSU Time Complexity : Adj List → O(E * union) ≅ O(E)

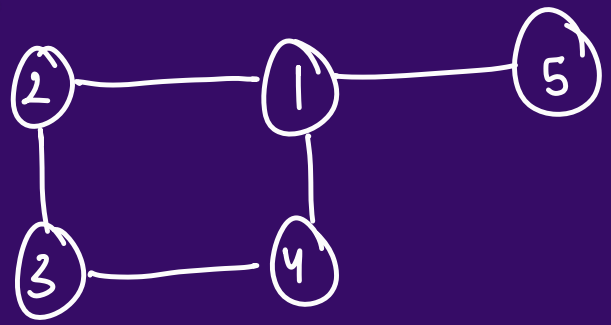
$\boxed{O(\alpha(n))}$ ~ O(4)

Ques:

Q : Redundant Connection

A tree is that graph where there are n-1 edges & no cycles.

n ~~are~~ nodes



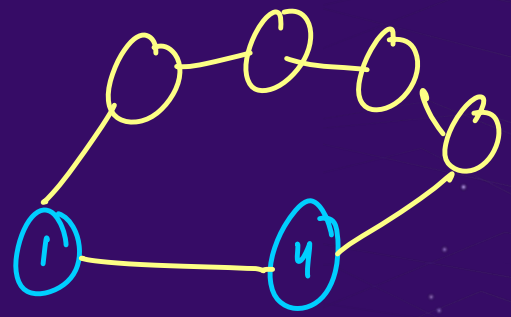
[Leetcode 684]

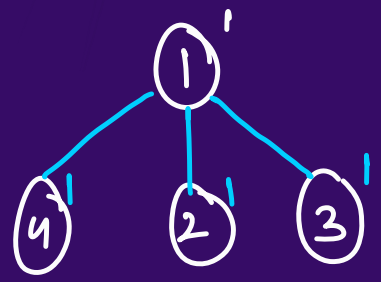
Ques:

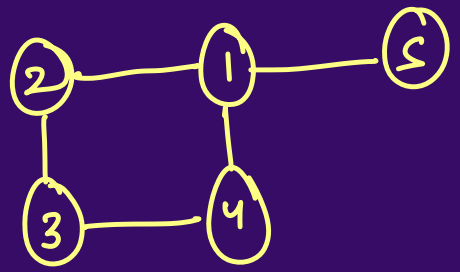
Q : Redundant Connection

cycle detected

edges = { {1, 2}, {2, 3}, {3, 4}, {1, 4}, {1, 5} }





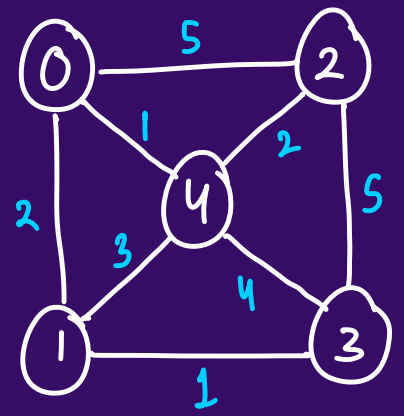


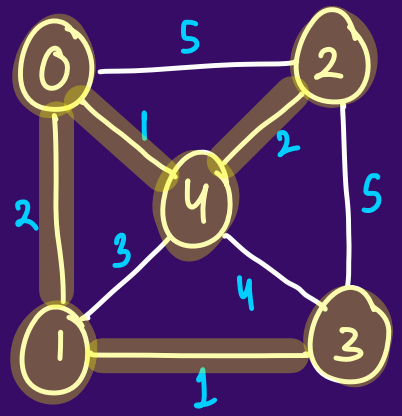
[Leetcode 684]

Kruskal's Algorithm

Minimum Spanning Tree

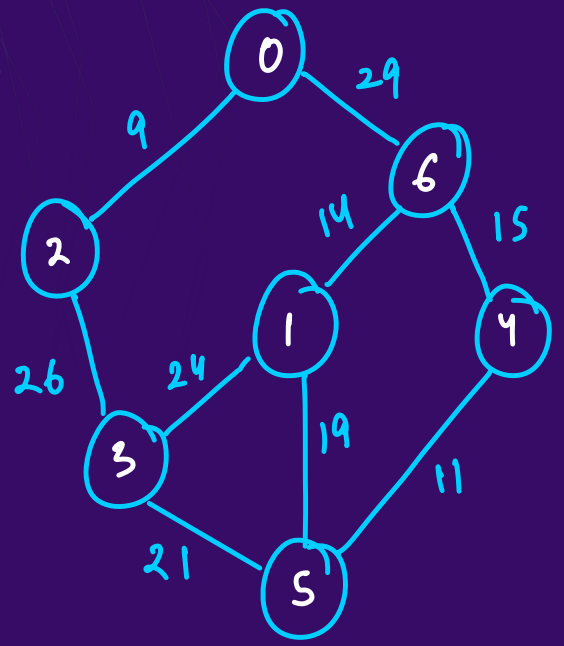
Prim's Algo.



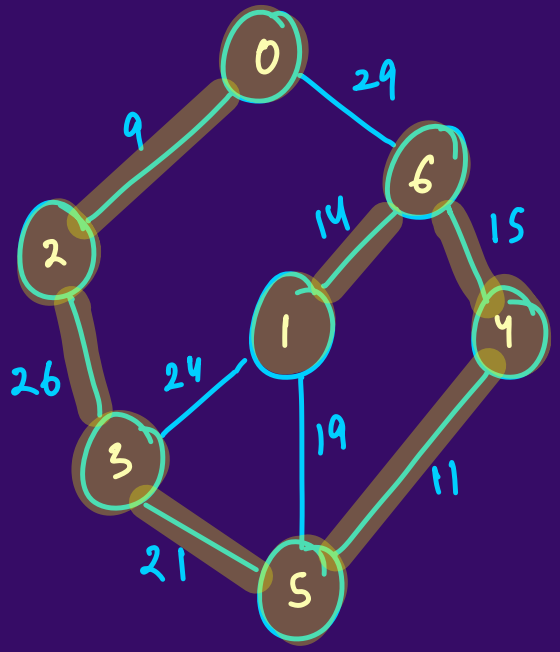


Kruskal's Algorithm

Minimum Spanning Tree



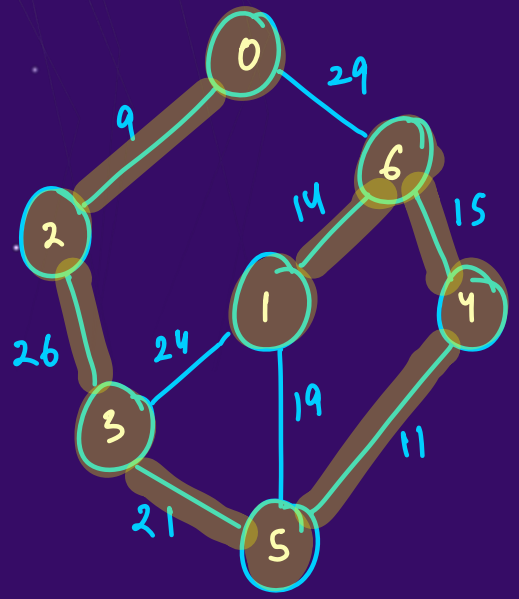
original



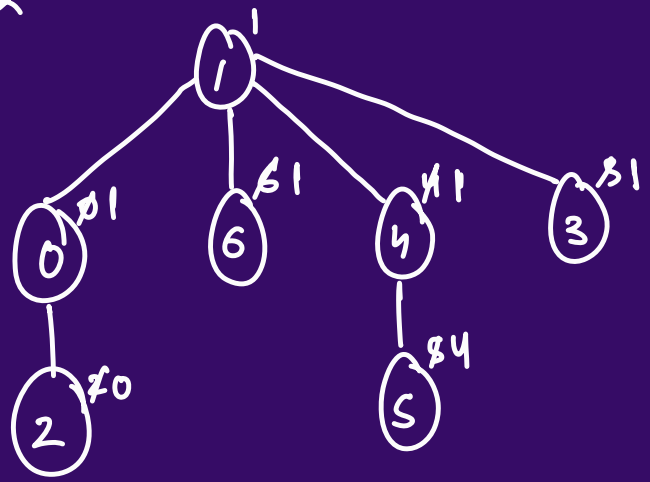
Kruskal

Kruskal's Algorithm

Minimum Spanning Tree



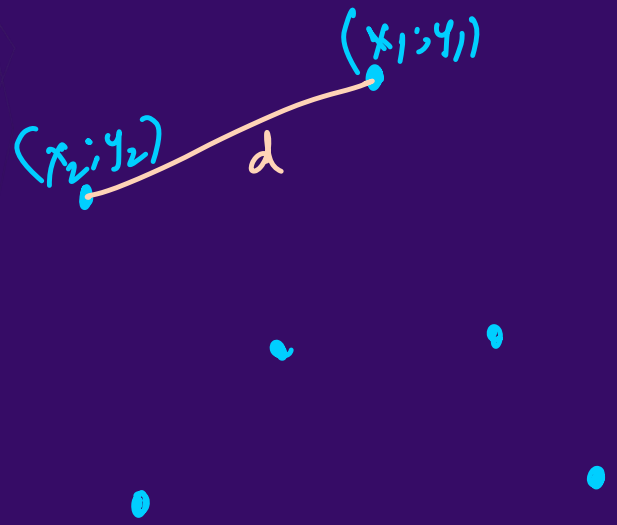
{2, 0, 9}, {5, 4, 11}, {1, 6, 14}, {6, 4, 15},
{1, 5, 19}, {3, 5, 21}, {3, 1, 24}, {2, 3, 26},
{0, 6, 29}



minCost = ~~0~~ 9 ~~20~~ ~~34~~ ~~49~~ ~~70~~
96

Ques:

Q : Min cost to connect all points



$$d = |x_1 - x_2| + |y_1 - y_2|$$

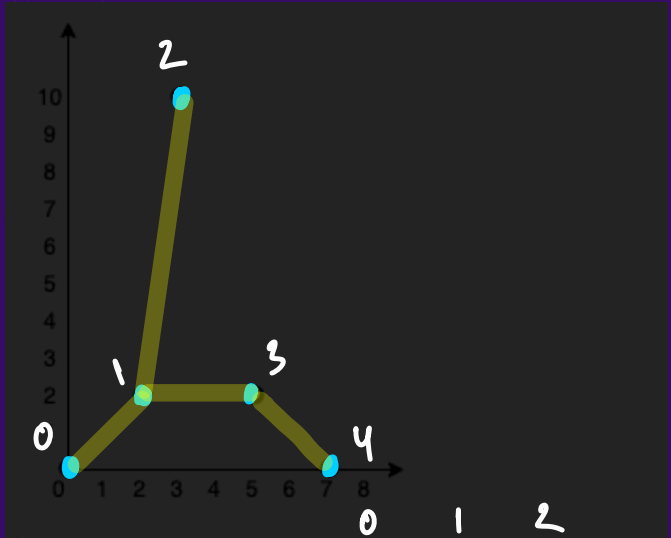
[Leetcode 1584]

Ques:

Q : Min cost to connect all points

indices of points

u, v, wt



0 1 2

3 4

```
Input: points = [[0,0],[2,2],[3,10],
[5,2],[7,0]]
```

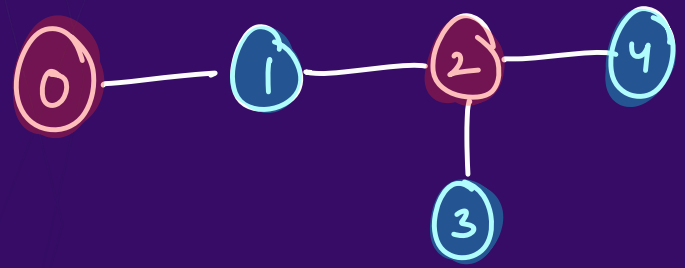
edgelist = {{0, 1, 4}, {0, 2, 13}, {0, 3, 7}, {0, 4, 7}, {1, 2, 9}, {1, 3, 3}, {1, 4, 9}, {2, 3, 10}, {2, 4, 14}, {3, 4, 4}}

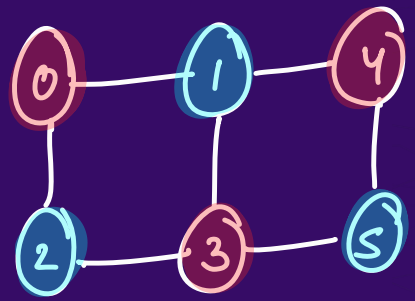
cost = 3 + 4 + 4

[Leetcode 1584]

Ques:

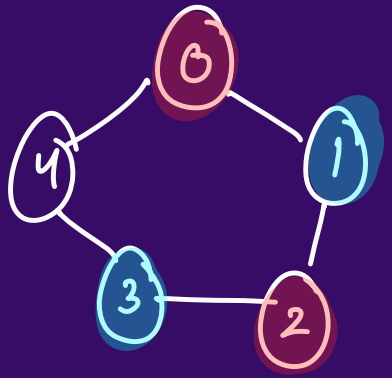
Q : Is Graph Bipartite 'Coloring of graph'





Non-cyclic

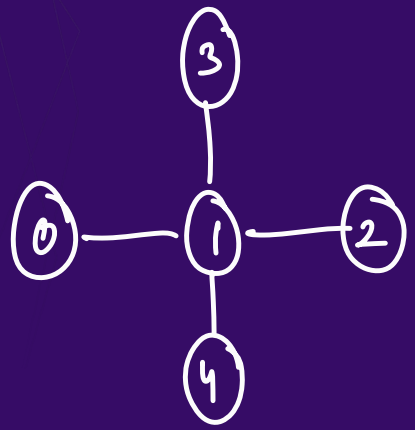
Even cyclic

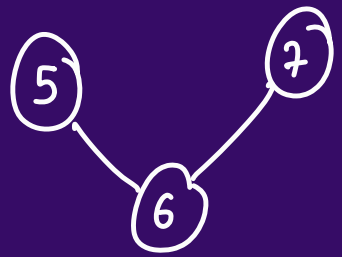


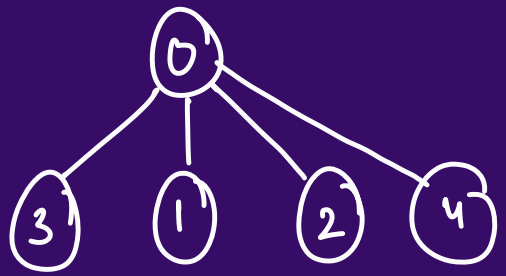
[Leetcode 785]

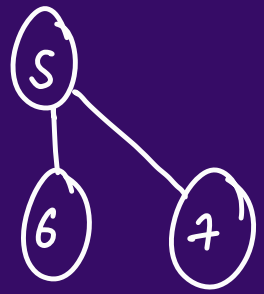
Ques:

Q : Is Graph Bipartite









[Leetcode 785]

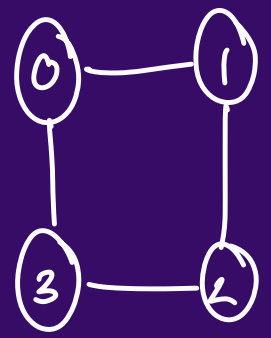
Ques:

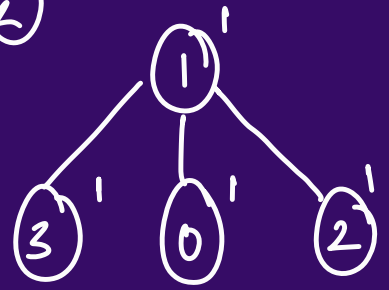
DSU → only detect cycles [odd/even]

Q : Is Graph Bipartite

cycle mili

(0, 1), (1, 2), (2, 3), (3, 0)

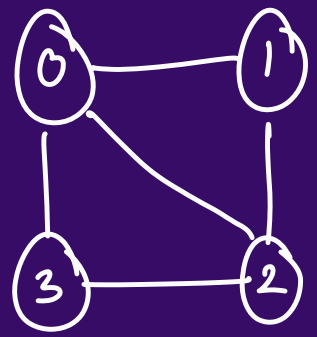


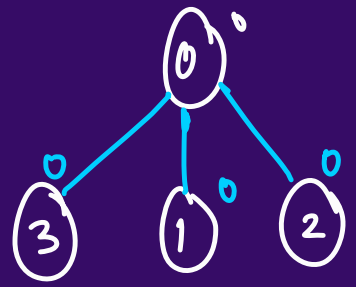


[Leetcode 785]

Ques:

Q : Is Graph Bipartite





cycle

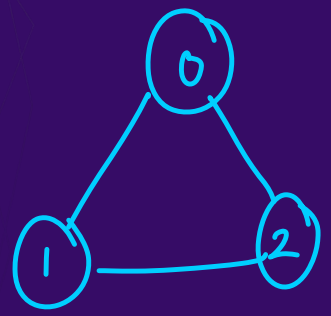
α α

(0, 1), (0, 2), (0, 3), (1, 2), (2, 3)

[Leetcode 785]

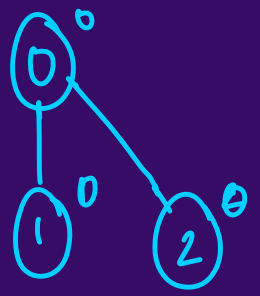
Ques:

Q : Is Graph Bipartite



(0, 1) (0, 2) (1, 2)

cycle



[Leetcode 785]

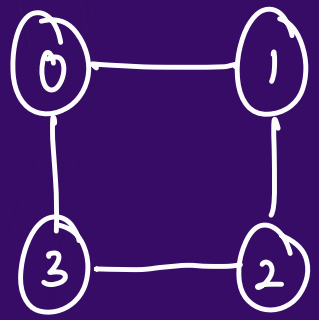
Ques:

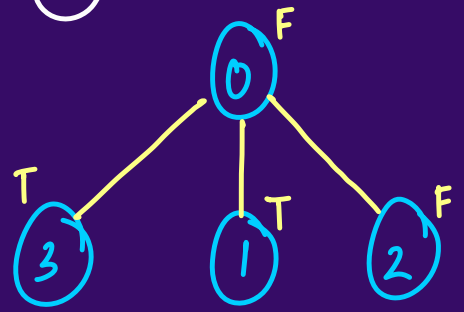
DSU → parent, size, parity → true (red), false (blue)
↓
color

Q : Is Graph Bipartite

(0, 1), (1, 2), (2, 3), (3, 0)

α → Cycle mili nai

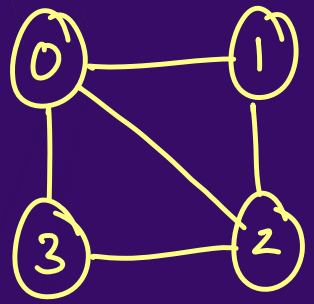


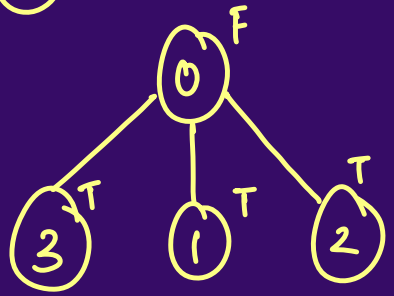


[Leetcode 785]

Ques:

Q : Is Graph Bipartite





Cycle

(0,1) (0,2) (0,3) (1,2) (2,3)

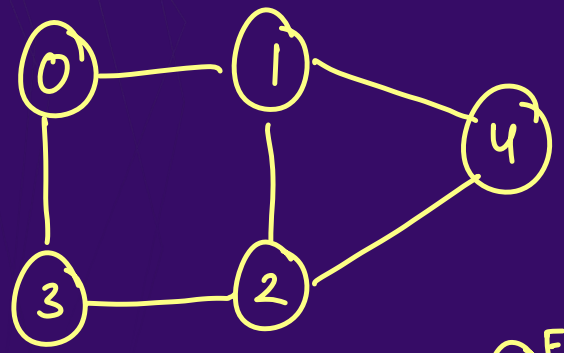
Same parity

odd cycle

[Leetcode 785]

Ques:

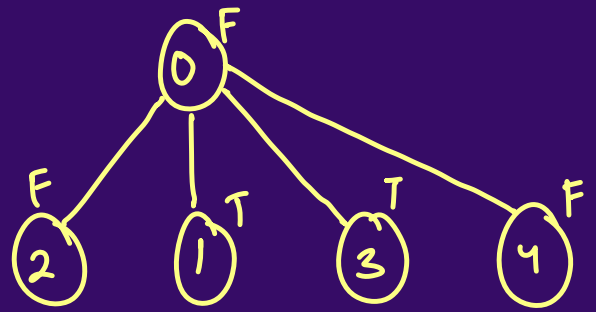
Q : Is Graph Bipartite



(0,1), (0,3) (1,2), (3,2) (1,4) (2,4)

(3,2) → diff parity (even)

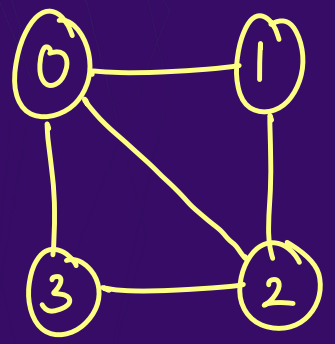
(2,4) → same parity (odd)



[Leetcode 785]

Ques:

Q : Is Graph Bipartite



0 : 3, 2, 1

1 : 0, 2

2 : 0, 3, 1

3 : 2, 0

leader
↓
find.

User defined data structure → DSU

[Leetcode 785]